



7662 - Can accretion explain the carbon dichotomy in planet-forming disks?

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	FM Tau	MIRI Medium Resolution Spectroscopy	(1) FM-Tau
	2	2MASS J15450887-3417333	MIRI Medium Resolution Spectroscopy	(2) 2MASS-J15450887-3417333

Folder	Observation	Label	Observing Template	Science Target
	3	2MASS J16001844-2230114	MIRI Medium Resolution Spectroscopy	(3) 2MASS-J16001844-2230114
	4	2MASS J16274978-2425219	MIRI Medium Resolution Spectroscopy	(4) 2MASS-J16274978-2425219
	5	2MASS J16081566-2222199	MIRI Medium Resolution Spectroscopy	(5) 2MASS-J16081566-2222199
	6	2MASS J16092697-3836269	MIRI Medium Resolution Spectroscopy	(6) 2MASS-J16092697-3836269
	7	2MASS J11025504-7721508	MIRI Medium Resolution Spectroscopy	(7) 2MASS-J11025504-7721508
	8	2MASS J16035793-1942108	MIRI Medium Resolution Spectroscopy	(8) 2MASS-J16035793-1942108
	9	2MASS J16041740-1942287	MIRI Medium Resolution Spectroscopy	(9) 2MASS-J16041740-1942287
	10	2MASS J16090002-1908368	MIRI Medium Resolution Spectroscopy	(10) 2MASS-J16090002-1908368
	11	2MASS J11432669-7804454	MIRI Medium Resolution Spectroscopy	(11) 2MASS-J11432669-7804454
	12	2MASS J16095361-1754474	MIRI Medium Resolution Spectroscopy	(12) 2MASS-J16095361-1754474
	13	2MASS J16063539-2516510	MIRI Medium Resolution Spectroscopy	(13) 2MASS-J16063539-2516510
	14	2MASS J15592523-4235066	MIRI Medium Resolution Spectroscopy	(14) 2MASS-J15592523-4235066

ABSTRACT

Planet-forming disks show a striking dichotomy in their chemical makeup: disks around low-mass stars often exhibit a higher carbon-to-oxygen ratio, while those around solar-mass stars display more oxygen-rich chemistry. These differences are believed to result from the sublimation and freeze-out of volatile species like CO, CO₂, and H₂O, but the precise origin of this carbon dichotomy remains uncertain. We hypothesize that the composition dichotomy in disks arises from the irreversible sublimation of carbon-bearing grains, and that the accretion rate, rather than stellar mass, plays a key role in driving carbon enrichment. Initial analyses of MIRI data published in the literature show a clear trend of higher C₂H₂ (relative to H₂O) columns with decreasing accretion rate, supporting our hypothesis, though much remains unexplored. To provide a stringent test, we propose to conduct a survey of 14 protoplanetary disks using MIRI-MRS, targeting systems across ranges of unexplored accretion rates and stellar masses, complemented by archival data from 8 low accretion sources. By measuring the relative amount of hydrocarbons like C₂H₂ and oxygen-rich species

like H₂O, we will determine whether low accretion rates indeed promote carbon-rich chemistries. To complement the observational analysis, we will perform a combination of thermochemical and dynamical modeling to investigate how accretion, pebble drift, and sublimation processes influence the molecular content of disks. This will offer new insights into how these factors shape disk chemistry, with direct implications for JWST observations of disks, and help us understand the composition of planets forming in these systems.

OBSERVING DESCRIPTION

The goal of this program is to observe MIRI-MRS spectra of 14 sources covering the full spectral range (4.9-28 micron) by using all three grating settings of the four MRS channels. Our sensitivities are adjusted to obtain a S/N~150 between 10-16 microns, where most of the molecular features of interest (C₂H₂, CO₂, HCN, C₆H₆, C₄H₂) are located. To minimize fringing, we also include target acquisition using the FND filter. Since the primary emission from these sources originates in the inner few au, and they are located at distances of 130-190 au, these sources will be unresolved point sources, therefore, we will use a 4-point dither pattern in the negative direction to increase the spatial and spectral sampling. The fast readout pattern is used for 9 of the sources where we expect fluxes higher than 15mJy at 12 micron (based on WISE W3 measurements). In the other 5 sources, we used the slow readout pattern to avoid exceeding the data volume limit.

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#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	FM-Tau	RA: 04 14 13.5842 (63.5566008d) Dec: +28 12 49.17 (28.21366d) Equinox: J2000	Proper Motion RA: 8.383 mas/yr Proper Motion Dec: -24.539999958506087 mas/yr Parallax: 0.0075753" Epoch of Position: 2000	
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i>				
<i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Protoplanetary disks]</i>				
(2)	2MASS-J15450887-3417333	RA: 15 45 8.8743 (236.2869763d) Dec: -34 17 33.45 (-34.29262d) Equinox: J2000	Proper Motion RA: -15.208 mas/yr Proper Motion Dec: -22.202000059223792 mas/yr Parallax: 0.0064595" Epoch of Position: 2000	
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i>				
<i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Protoplanetary disks]</i>				
(3)	2MASS-J16001844-2230114	RA: 16 00 18.4309 (240.0767954d) Dec: -22 30 11.54 (-22.50321d) Equinox: J2000	Proper Motion RA: -6.673 mas/yr Proper Motion Dec: -17.06299997294991 mas/yr Parallax: 0.0061605" Epoch of Position: 2000	
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i>				
<i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Protoplanetary disks]</i>				
(4)	2MASS-J16274978-2425219	RA: 16 27 49.7891 (246.9574546d) Dec: -24 25 21.93 (-24.42276d) Equinox: J2000	Proper Motion RA: -9.534 mas/yr Proper Motion Dec: -21.268999898893526 mas/yr Parallax: 0.0048541" Epoch of Position: 2000	
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i>				
<i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Protoplanetary disks]</i>				
(5)	2MASS-J16081566-2222199	RA: 16 08 15.6663 (242.0652763d) Dec: -22 22 20.05 (-22.37224d) Equinox: J2000	Proper Motion RA: -10.436 mas/yr Proper Motion Dec: -24.561999998695683 mas/yr Parallax: 0.0071627" Epoch of Position: 2000	
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i>				
<i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Protoplanetary disks]</i>				

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(6)	2MASS-J16092697-3836269	RA: 16 09 26.9796 (242.3624150d) Dec: -38 36 26.93 (-38.60748d) Equinox: J2000	Proper Motion RA: -9.252 mas/yr Proper Motion Dec: -22.9470000249421 mas/yr Parallax: 0.006281699999999994" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> Category=Star Description=[Protoplanetary disks]			
(7)	2MASS-J11025504-7721508	RA: 11 02 55.0256 (165.7292733d) Dec: -77 21 50.81 (-77.36411d) Equinox: J2000	Proper Motion RA: -23.791 mas/yr Proper Motion Dec: 1.905 mas/yr Parallax: 0.0054288" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> Category=Star Description=[Protoplanetary disks]			
(8)	2MASS-J16035793-1942108	RA: 16 03 57.9355 (240.9913979d) Dec: -19 42 10.91 (-19.70303d) Equinox: J2000	Proper Motion RA: -11.27 mas/yr Proper Motion Dec: -22.09600008882262 mas/yr Parallax: 0.0064857" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> Category=Star Description=[Protoplanetary disks]			
(9)	2MASS-J16041740-1942287	RA: 16 04 17.4053 (241.0725221d) Dec: -19 42 28.82 (-19.70801d) Equinox: J2000	Proper Motion RA: -9.859 mas/yr Proper Motion Dec: -21.64599993648153 mas/yr Parallax: 0.0064892000000000005" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> Category=Star Description=[Protoplanetary disks]			
(10)	2MASS-J16090002-1908368	RA: 16 09 0.0228 (242.2500950d) Dec: -19 08 36.87 (-19.14357d) Equinox: J2000	Proper Motion RA: -9.439 mas/yr Proper Motion Dec: -24.890999929993995 mas/yr Parallax: 0.0073027" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> Category=Star Description=[Protoplanetary disks]			

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(11)	2MASS-J11432669-7804454	RA: 11 43 26.6961 (175.8612337d) Dec: -78 04 45.49 (-78.07930d) Equinox: J2000	Proper Motion RA: -44.597 mas/yr Proper Motion Dec: -5.044999988967902 mas/yr Parallax: 0.006517400000000004" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> Category=Star Description=[Protoplanetary disks]			
(12)	2MASS-J16095361-1754474	RA: 16 09 53.6173 (242.4734054d) Dec: -17 54 47.43 (-17.91318d) Equinox: J2000	Proper Motion RA: -9.67 mas/yr Proper Motion Dec: -21.158999902581854 mas/yr Parallax: 0.0063049" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> Category=Star Description=[Protoplanetary disks]			
(13)	2MASS-J16063539-2516510	RA: 16 06 35.3930 (241.6474708d) Dec: -25 16 51.20 (-25.28089d) Equinox: J2000	Proper Motion RA: -10.95 mas/yr Proper Motion Dec: -25.884000069709145 mas/yr Parallax: 0.007207900000000001" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> Category=Star Description=[Protoplanetary disks]			
(14)	2MASS-J15592523-4235066	RA: 15 59 25.2340 (239.8551417d) Dec: -42 35 6.57 (-42.58516d) Equinox: J2000	Proper Motion RA: -17.205 mas/yr Proper Motion Dec: -26.566999940769165 mas/yr Parallax: 0.006791" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> Category=Star Description=[Protoplanetary disks]			

Proposal 7662 - Observation 1 - Can accretion explain the carbon dichotomy in planet-forming disks?

Observation	Proposal 7662, Observation 1: FM Tau													Sat Apr 04 00:00:12 GMT 2026
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(1)	FM-Tau	RA: 04 14 13.5842 (63.5566008d) Dec: +28 12 49.17 (28.21366d) Equinox: J2000				Proper Motion RA: 8.383 mas/yr Proper Motion Dec: -24.539999958506087 mas/yr Parallax: 0.0075753" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Star Description=[Protoplanetary disks]													
Acquisition	#	Target	Filter	Readout Pattern		Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	Optional ETC ID			
	1	SAME	FND	FAST		4	1		1	11.1	224573			
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction			
	All MRS		YES				FULL				Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F770W	FASTR1	20	1	1	Dither 1	4	4	222.003		
	1	SHORT(A)	MRSLONG		FASTR1	20	1	1	Dither 1	4	4	222.003		
	1	SHORT(A)	MRSSHORT		FASTR1	20	1	1	Dither 1	4	4	222.003		
	2		IMAGER	F770W	FASTR1	20	1	1	Dither 1	4	4	222.003		
	2	MEDIUM(B)	MRSLONG		FASTR1	20	1	1	Dither 1	4	4	222.003		
	2	MEDIUM(B)	MRSSHORT		FASTR1	20	1	1	Dither 1	4	4	222.003		
	3		IMAGER	F770W	FASTR1	20	1	1	Dither 1	4	4	222.003		
	3	LONG(C)	MRSLONG		FASTR1	20	1	1	Dither 1	4	4	222.003		
	3	LONG(C)	MRSSHORT		FASTR1	20	1	1	Dither 1	4	4	222.003		

Proposal 7662 - Observation 2 - Can accretion explain the carbon dichotomy in planet-forming disks?

Observation	Proposal 7662, Observation 2: 2MASS J15450887-3417333												Sat Apr 04 00:00:12 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(2)	2MASS-J15450887-3417333	RA: 15 45 8.8743 (236.2869763d) Dec: -34 17 33.45 (-34.29262d) Equinox: J2000				Proper Motion RA: -15.208 mas/yr Proper Motion Dec: -22.202000059223792 mas/yr Parallax: 0.0064595" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Protoplanetary disks]												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	Optional ETC ID			
	1	SAME	FND	FAST	4	1		1	11.1	224573			
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction		
	All MRS		YES				FULL				Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	25	1	1	Dither 1	4	4	277.504	
	1	SHORT(A)	MRSLONG		FASTR1	25	1	1	Dither 1	4	4	277.504	
	1	SHORT(A)	MRSSHORT		FASTR1	25	1	1	Dither 1	4	4	277.504	
	2		IMAGER	F770W	FASTR1	25	1	1	Dither 1	4	4	277.504	
	2	MEDIUM(B)	MRSLONG		FASTR1	25	1	1	Dither 1	4	4	277.504	
	2	MEDIUM(B)	MRSSHORT		FASTR1	25	1	1	Dither 1	4	4	277.504	
	3		IMAGER	F770W	FASTR1	25	1	1	Dither 1	4	4	277.504	
	3	LONG(C)	MRSLONG		FASTR1	25	1	1	Dither 1	4	4	277.504	
	3	LONG(C)	MRSSHORT		FASTR1	25	1	1	Dither 1	4	4	277.504	

Proposal 7662 - Observation 3 - Can accretion explain the carbon dichotomy in planet-forming disks?

Observation	Proposal 7662, Observation 3: 2MASS J16001844-2230114												Sat Apr 04 00:00:12 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(3)	2MASS-J16001844-2230114	RA: 16 00 18.4309 (240.0767954d) Dec: -22 30 11.54 (-22.50321d) Equinox: J2000				Proper Motion RA: -6.673 mas/yr Proper Motion Dec: -17.06299997294991 mas/yr Parallax: 0.0061605" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Protoplanetary disks]												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	FND	FAST	4	1	1	11.1	224573				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray			Grating Wheel Direction			
	All MRS		YES				FULL			Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005	
	1	SHORT(A)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	
	1	SHORT(A)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	
	2		IMAGER	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005	
	2	MEDIUM(B)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	
	2	MEDIUM(B)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	
	3		IMAGER	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005	
	3	LONG(C)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	
	3	LONG(C)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	

Proposal 7662 - Observation 4 - Can accretion explain the carbon dichotomy in planet-forming disks?

Observation	Proposal 7662, Observation 4: 2MASS J16274978-2425219												Sat Apr 04 00:00:12 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(4)	2MASS-J16274978-2425219	RA: 16 27 49.7891 (246.9574546d) Dec: -24 25 21.93 (-24.42276d) Equinox: J2000				Proper Motion RA: -9.534 mas/yr Proper Motion Dec: -21.268999898893526 mas/yr Parallax: 0.0048541" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Protoplanetary disks]												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	Optional ETC ID			
	1	SAME	FND	FAST	4	1		1	11.1	224573			
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction		
	All MRS		YES				FULL				Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005	
	1	SHORT(A)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	
	1	SHORT(A)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	
	2		IMAGER	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005	
	2	MEDIUM(B)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	
	2	MEDIUM(B)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	
	3		IMAGER	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005	
	3	LONG(C)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	
	3	LONG(C)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	

Proposal 7662 - Observation 5 - Can accretion explain the carbon dichotomy in planet-forming disks?

Observation	Proposal 7662, Observation 5: 2MASS J16081566-2222199												Sat Apr 04 00:00:12 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(5)	2MASS-J16081566-2222199	RA: 16 08 15.6663 (242.0652763d) Dec: -22 22 20.05 (-22.37224d) Equinox: J2000				Proper Motion RA: -10.436 mas/yr Proper Motion Dec: -24.561999998695683 mas/yr Parallax: 0.0071627" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Protoplanetary disks]												
Acquisition	#	Target	Filter	Readout Pattern		Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	Optional ETC ID		
	1	SAME	FND	FAST		4	1		1	11.1	224573		
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction		
	All MRS		YES				FULL				Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005	
	1	SHORT(A)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	
	1	SHORT(A)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	
	2		IMAGER	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005	
	2	MEDIUM(B)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	
	2	MEDIUM(B)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	
	3		IMAGER	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005	
	3	LONG(C)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	
	3	LONG(C)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	

Proposal 7662 - Observation 6 - Can accretion explain the carbon dichotomy in planet-forming disks?

Observation	Proposal 7662, Observation 6: 2MASS J16092697-3836269												Sat Apr 04 00:00:12 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(6)	2MASS-J16092697-3836269	RA: 16 09 26.9796 (242.3624150d) Dec: -38 36 26.93 (-38.60748d) Equinox: J2000				Proper Motion RA: -9.252 mas/yr Proper Motion Dec: -22.9470000249421 mas/yr Parallax: 0.006281699999999994" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Protoplanetary disks]												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations		Total Exposure Time		Optional ETC ID	
	1	SAME	FND	FAST	8	1		1		22.2		224573	
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction		
	All MRS		YES				FULL				Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008	
	1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
	1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
	2		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008	
	2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
	3		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008	
	3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
	3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	

Proposal 7662 - Observation 7 - Can accretion explain the carbon dichotomy in planet-forming disks?

Observation	Proposal 7662, Observation 7: 2MASS J11025504-7721508												Sat Apr 04 00:00:12 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(7)	2MASS-J11025504-7721508	RA: 11 02 55.0256 (165.7292733d) Dec: -77 21 50.81 (-77.36411d) Equinox: J2000				Proper Motion RA: -23.791 mas/yr Proper Motion Dec: 1.905 mas/yr Parallax: 0.0054288" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Protoplanetary disks]												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	FND	FAST	8	1	1	22.2	224573				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction		
	All MRS		YES				FULL				Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008	
	1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
	1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
	2		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008	
	2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
	3		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008	
	3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
	3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	

Proposal 7662 - Observation 8 - Can accretion explain the carbon dichotomy in planet-forming disks?

Observation	Proposal 7662, Observation 8: 2MASS J16035793-1942108												Sat Apr 04 00:00:12 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(8)	2MASS-J16035793-1942108	RA: 16 03 57.9355 (240.9913979d) Dec: -19 42 10.91 (-19.70303d) Equinox: J2000				Proper Motion RA: -11.27 mas/yr Proper Motion Dec: -22.09600008882262 mas/yr Parallax: 0.0064857" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Protoplanetary disks]												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations		Total Exposure Time		Optional ETC ID	
	1	SAME	FND	FAST	8	1		1		22.2		224573	
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction		
	All MRS		YES				FULL				Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008	
	1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
	1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
	2		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008	
	2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
	3		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008	
	3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
	3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	

Proposal 7662 - Observation 9 - Can accretion explain the carbon dichotomy in planet-forming disks?

Observation	Proposal 7662, Observation 9: 2MASS J16041740-1942287												Sat Apr 04 00:00:12 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(9)	2MASS-J16041740-1942287	RA: 16 04 17.4053 (241.0725221d) Dec: -19 42 28.82 (-19.70801d) Equinox: J2000				Proper Motion RA: -9.859 mas/yr Proper Motion Dec: -21.64599993648153 mas/yr Parallax: 0.0064892000000000005" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Protoplanetary disks]												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations		Total Exposure Time		Optional ETC ID	
	1	SAME	FND	FAST	8	1		1		22.2		224573	
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction		
	All MRS		YES				FULL				Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008	
	1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
	1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
	2		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008	
	2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
	3		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008	
	3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
	3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	

Proposal 7662 - Observation 10 - Can accretion explain the carbon dichotomy in planet-forming disks?

Observation	Proposal 7662, Observation 10: 2MASS J16090002-1908368												Sat Apr 04 00:00:12 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 10:1) Warning (Form): Data Excess over lower threshold												
	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(10)	2MASS-J16090002-1908368	RA: 16 09 0.0228 (242.2500950d) Dec: -19 08 36.87 (-19.14357d) Equinox: J2000				Proper Motion RA: -9.439 mas/yr Proper Motion Dec: -24.890999929993995 mas/yr Parallax: 0.0073027" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Protoplanetary disks]												
Acquisition	#	Target	Filter	Readout Pattern		Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	Optional ETC ID		
	1	SAME	FND	FAST		10	1		1	27.75	224573		
Template	Primary Channel		Simultaneous Imaging				Imager Subarray			Grating Wheel Direction			
	All MRS		YES				FULL			Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	57	3	1	Dither 1	4	12	1920.328	
	1	SHORT(A)	MRSLONG		FASTR1	58	3	1	Dither 1	4	12	1953.628	
	1	SHORT(A)	MRSSHORT		FASTR1	57	3	1	Dither 1	4	12	1920.328	
	2		IMAGER	F770W	FASTR1	57	3	1	Dither 1	4	12	1920.328	
	2	MEDIUM(B)	MRSLONG		FASTR1	58	3	1	Dither 1	4	12	1953.628	
	2	MEDIUM(B)	MRSSHORT		FASTR1	57	3	1	Dither 1	4	12	1920.328	
	3		IMAGER	F770W	FASTR1	57	3	1	Dither 1	4	12	1920.328	
	3	LONG(C)	MRSLONG		FASTR1	58	3	1	Dither 1	4	12	1953.628	
	3	LONG(C)	MRSSHORT		FASTR1	57	3	1	Dither 1	4	12	1920.328	

Proposal 7662 - Observation 11 - Can accretion explain the carbon dichotomy in planet-forming disks?

Observation	Proposal 7662, Observation 11: 2MASS J11432669-7804454												Sat Apr 04 00:00:12 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 11:1) Warning (Form): Data Excess over lower threshold												
	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(11)	2MASS-J11432669-7804454	RA: 11 43 26.6961 (175.8612337d) Dec: -78 04 45.49 (-78.07930d) Equinox: J2000				Proper Motion RA: -44.597 mas/yr Proper Motion Dec: -5.044999988967902 mas/yr Parallax: 0.006517400000000004" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Protoplanetary disks]												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	FND	FAST	10	1	1	27.75	224573				
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS		YES			FULL			Allow Auto Reorder				
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	57	3	1	Dither 1	4	12	1920.328	
	1	SHORT(A)	MRSLONG		FASTR1	58	3	1	Dither 1	4	12	1953.628	
	1	SHORT(A)	MRSSHORT		FASTR1	57	3	1	Dither 1	4	12	1920.328	
	2		IMAGER	F770W	FASTR1	57	3	1	Dither 1	4	12	1920.328	
	2	MEDIUM(B)	MRSLONG		FASTR1	58	3	1	Dither 1	4	12	1953.628	
	2	MEDIUM(B)	MRSSHORT		FASTR1	57	3	1	Dither 1	4	12	1920.328	
	3		IMAGER	F770W	FASTR1	57	3	1	Dither 1	4	12	1920.328	
	3	LONG(C)	MRSLONG		FASTR1	58	3	1	Dither 1	4	12	1953.628	
	3	LONG(C)	MRSSHORT		FASTR1	57	3	1	Dither 1	4	12	1920.328	

Proposal 7662 - Observation 12 - Can accretion explain the carbon dichotomy in planet-forming disks?

Observation	Proposal 7662, Observation 12: 2MASS J16095361-1754474												Sat Apr 04 00:00:12 GMT 2026
	Diagnostic Status: Warning												
Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(Visit 12:1) Warning (Form): Data Excess over lower threshold												
	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(12)	2MASS-J16095361-1754474	RA: 16 09 53.6173 (242.4734054d) Dec: -17 54 47.43 (-17.91318d) Equinox: J2000				Proper Motion RA: -9.67 mas/yr Proper Motion Dec: -21.158999902581854 mas/yr Parallax: 0.0063049" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Protoplanetary disks]												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations		Total Exposure Time		Optional ETC ID	
	1	SAME	FND	FAST	10	1		1		27.75		224573	
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction		
	All MRS		YES				FULL				Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	72	3	1	Dither 1	4	12	2419.835	
	1	SHORT(A)	MRSLONG		FASTR1	72	3	1	Dither 1	4	12	2419.835	
	1	SHORT(A)	MRSSHORT		FASTR1	71	3	1	Dither 1	4	12	2386.534	
	2		IMAGER	F770W	FASTR1	72	3	1	Dither 1	4	12	2419.835	
	2	MEDIUM(B)	MRSLONG		FASTR1	72	3	1	Dither 1	4	12	2419.835	
	2	MEDIUM(B)	MRSSHORT		FASTR1	71	3	1	Dither 1	4	12	2386.534	
	3		IMAGER	F770W	FASTR1	72	3	1	Dither 1	4	12	2419.835	
	3	LONG(C)	MRSLONG		FASTR1	72	3	1	Dither 1	4	12	2419.835	
	3	LONG(C)	MRSSHORT		FASTR1	71	3	1	Dither 1	4	12	2386.534	

Proposal 7662 - Observation 13 - Can accretion explain the carbon dichotomy in planet-forming disks?

Observation	Proposal 7662, Observation 13: 2MASS J16063539-2516510												Sat Apr 04 00:00:12 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 13:1) Warning (Form): Data Excess over lower threshold												
	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(13)	2MASS-J16063539-2516510	RA: 16 06 35.3930 (241.6474708d) Dec: -25 16 51.20 (-25.28089d) Equinox: J2000				Proper Motion RA: -10.95 mas/yr Proper Motion Dec: -25.884000069709145 mas/yr Parallax: 0.007207900000000001" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Protoplanetary disks]												
Acquisition	#	Target	Filter	Readout Pattern		Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	Optional ETC ID		
	1	SAME	FND	FAST		10	1		1	27.75	224573		
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction		
	All MRS		YES				FULL				Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	72	3	1	Dither 1	4	12	2419.835	
	1	SHORT(A)	MRSLONG		FASTR1	72	3	1	Dither 1	4	12	2419.835	
	1	SHORT(A)	MRSSHORT		FASTR1	71	3	1	Dither 1	4	12	2386.534	
	2		IMAGER	F770W	FASTR1	72	3	1	Dither 1	4	12	2419.835	
	2	MEDIUM(B)	MRSLONG		FASTR1	72	3	1	Dither 1	4	12	2419.835	
	2	MEDIUM(B)	MRSSHORT		FASTR1	71	3	1	Dither 1	4	12	2386.534	
	3		IMAGER	F770W	FASTR1	72	3	1	Dither 1	4	12	2419.835	
	3	LONG(C)	MRSLONG		FASTR1	72	3	1	Dither 1	4	12	2419.835	
	3	LONG(C)	MRSSHORT		FASTR1	71	3	1	Dither 1	4	12	2386.534	

Proposal 7662 - Observation 14 - Can accretion explain the carbon dichotomy in planet-forming disks?

Observation	Proposal 7662, Observation 14: 2MASS J15592523-4235066												Sat Apr 04 00:00:12 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 14:1) Warning (Form): Data Excess over lower threshold												
	(Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(14)	2MASS-J15592523-4235066	RA: 15 59 25.2340 (239.8551417d) Dec: -42 35 6.57 (-42.58516d) Equinox: J2000				Proper Motion RA: -17.205 mas/yr Proper Motion Dec: -26.566999940769165 mas/yr Parallax: 0.006791" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Protoplanetary disks]												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	FND	FAST	10	1	1	27.75	224573				
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS		YES			FULL			Allow Auto Reorder				
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	72	3	1	Dither 1	4	12	2419.835	
	1	SHORT(A)	MRSLONG		FASTR1	72	3	1	Dither 1	4	12	2419.835	
	1	SHORT(A)	MRSSHORT		FASTR1	71	3	1	Dither 1	4	12	2386.534	
	2		IMAGER	F770W	FASTR1	72	3	1	Dither 1	4	12	2419.835	
	2	MEDIUM(B)	MRSLONG		FASTR1	72	3	1	Dither 1	4	12	2419.835	
	2	MEDIUM(B)	MRSSHORT		FASTR1	71	3	1	Dither 1	4	12	2386.534	
	3		IMAGER	F770W	FASTR1	72	3	1	Dither 1	4	12	2419.835	
	3	LONG(C)	MRSLONG		FASTR1	72	3	1	Dither 1	4	12	2419.835	
	3	LONG(C)	MRSSHORT		FASTR1	71	3	1	Dither 1	4	12	2386.534	